

MIRAMICHI, NB, Canada

WMO: 717440

Lat: 47.0094N

Lon: 65.4650W

Elev: 34

StdP: 100.92

Time Zone: -4.00 (NAA)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -23.6	(c) -20.9	(d) -28.6	(e) 0.3	(f) -22.4	(g) -26.0	(h) 0.4	(i) -19.9	(j) 12.1	(k) 0.1	(l) 10.4	(m) -2.6	(n) 2.4	(o) 260	(p) 0.617

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.7	30.1	20.7	28.2	19.8	26.4	18.8	22.3	27.5	21.3	25.8	20.3	24.5	5.2	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.7	15.4	24.6	19.7	14.5	23.6	18.8	13.7	22.7	65.8	27.6	62.1	26.0	58.7	24.5	29.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 10.0	(b) 8.7	(c) 7.6	(e) -27.9	(f) 33.4	(g) 2.5	(h) 1.8	(i) -29.7	(j) 34.7	(k) -31.1	(l) 35.7	(m) -32.5	(n) 36.7	(o) -34.3	(p) 38.1		
(a) 10.0	(b) 8.7	(c) 7.6	(e) -28.1	(f) 24.5	(g) 2.4	(h) 1.4	(i) -29.9	(j) 25.5	(k) -31.3	(l) 26.3	(m) -32.7	(n) 27.1	(o) -34.4	(p) 28.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	5.7	-9.1	-8.3	-3.0	3.6	10.3	16.0	20.0	19.0	14.6	7.9	1.5	-5.1		
	DBStd	11.08	6.27	6.02	5.30	3.88	4.11	3.99	2.95	2.99	3.85	4.04	4.92	5.53		
	HDD10.0	2575	592	512	404	197	46	3	0	0	5	90	259	468		
	HDD18.3	4766	851	745	662	443	251	92	16	27	124	323	506	726		
	CDD10.0	1003	0	0	1	3	56	183	310	279	142	26	3	0		
	CDD18.3	152	0	0	0	0	3	22	68	48	11	1	0	0		
	CDH23.3	1316	0	0	1	3	57	242	542	382	87	3	0	0		
	CDH26.7	344	0	0	0	1	12	70	142	99	21	0	0	0		
(14) Wind	WSAvg	3.4	3.4	3.6	3.8	3.7	3.6	3.4	3.1	3.0	3.1	3.4	3.5	3.5		
(15) Precipitation	PrecAvg	966	77	65	74	64	89	79	88	82	82	102	86	82		
	PrecMax	1229	134	130	155	141	161	146	190	175	202	181	151	211		
	PrecMin	675	13	4	17	23	15	29	10	22	20	20	34	13		
	PrecStd	117	32	31	34	30	35	27	42	39	44	44	31	38		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.1	9.2	13.9	20.8	28.4	31.7	32.2	32.2	29.6	23.1	17.7	10.3		
		MCWB	9.3	6.8	7.8	11.4	16.5	20.8	21.8	21.5	21.0	16.9	16.0	9.5		
	2%	DB	5.5	4.8	8.6	15.6	24.6	28.7	30.0	29.2	25.7	19.2	14.6	6.9		
		MCWB	4.4	2.8	4.6	9.6	15.1	19.0	21.1	20.5	19.4	14.9	12.6	6.0		
	5%	DB	2.6	2.4	6.3	12.9	21.2	26.0	28.1	27.2	23.2	16.8	11.5	4.2		
		MCWB	1.4	0.8	3.1	7.8	13.7	17.6	20.1	19.7	17.7	13.4	9.4	2.9		
	10%	DB	0.1	0.7	4.2	10.4	18.4	23.8	26.4	25.3	21.3	14.5	8.6	2.1		
		MCWB	-0.8	-0.5	1.6	6.0	12.2	16.6	19.1	18.7	17.0	11.6	6.8	1.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.4	7.5	9.7	14.3	19.0	23.0	23.9	23.4	22.4	18.0	16.3	9.6		
		MCDB	10.0	9.1	12.1	18.5	25.0	30.1	29.6	28.7	27.3	20.8	17.6	10.4		
	2%	WB	4.6	3.2	5.6	11.1	16.6	20.8	22.5	22.0	20.7	15.8	13.1	6.1		
		MCDB	5.3	4.5	7.8	13.7	21.4	25.9	27.5	27.1	23.9	18.1	14.2	6.8		
	5%	WB	1.4	1.0	3.5	8.5	14.8	19.4	21.4	20.9	19.2	13.8	9.8	3.1		
		MCDB	2.4	2.1	5.6	11.9	19.5	23.8	26.1	25.1	22.2	16.3	11.1	3.9		
	10%	WB	-0.7	-0.5	1.9	6.4	12.9	17.9	20.4	19.9	17.7	12.0	7.0	1.2		
		MCDB	0.0	0.6	3.9	10.0	17.9	21.9	24.8	23.8	20.4	14.2	8.5	1.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.2	10.7	9.9	9.6	11.3	11.2	10.7	10.8	10.9	9.4	7.7	8.1		
		MCDBR	12.5	11.0	12.1	14.6	17.1	15.4	13.3	13.6	13.2	12.6	11.2	8.7		
	5% WB	MCWBR	12.3	9.5	9.0	9.6	9.5	8.0	6.4	6.8	7.8	8.8	9.9	8.2		
		MCWBR	12.4	10.0	10.7	12.9	14.6	12.5	11.2	11.1	11.0	10.8	10.5	8.5		
(40) Clear-Sky Solar Irradiance	taub	taub	0.283	0.281	0.317	0.357	0.369	0.391	0.417	0.392	0.360	0.328	0.300	0.281		
		taud	2.405	2.379	2.390	2.381	2.404	2.338	2.305	2.366	2.443	2.526	2.552	2.487		
	Ebn at Noon	Ebn at Noon	812	896	909	898	896	874	844	852	851	832	796	779		
		Edh at Noon	69	86	100	112	113	122	125	113	95	75	60	57		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.34	2.12	3.33	4.25	5.06	5.45	5.52	4.83	3.80	2.38	1.30	0.98		
	RadStd	RadStd	0.14	0.23	0.23	0.39	0.50	0.41	0.39	0.32	0.33	0.21	0.16	0.10		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (31 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page